# Joseph T. Iosue

*Doctoral Candidate*

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*Mathematical Physics*
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## Education

2020 – **University of Maryland**, *Doctor of Philosophy (Physics)*, JQI Graduate Fellow
Advisors: Prof. Alexey Gorshkov and Prof. Victor Albert

2015 – 2019 **Massachusetts Institute of Technology**, *Bachelor of Science*
B.S. (Physics), Minor (Computer Science) – GPA 4.9/5.0

## Research Experience

2020 – **University of Maryland, Department of Physics**, *Research Assistant*
Affiliations:
- Joint Quantum Institute (JQI)
- Joint Center for Quantum Information and Computer Science (QuICS)

2019 – 2020 **QC Ware, Corp**, *Quantum Algorithms Researcher*
Developed software and algorithms for customer use cases, including [1].

2017 – 2018 **MIT Laboratory for Nuclear Science**, *Undergraduate Researcher*
Supervisor: Prof. Or Hen. Studied proton/neutron dynamics in asymmetric nuclei with C++ and ROOT.

2018 **Los Alamos National Laboratory**, *Quantum Computing Summer Fellowship*
Supervisor: Dr. Patrick Coles. Developed and published a novel quantum algorithm [3].

2017 **Joint Quantum Institute, University of Maryland**, *Summer Researcher*
Supervisor: Prof. Alexey Gorshkov. Studied quantum phase transitions via quench dynamics [2].

2016 **MIT Plasma Science and Fusion Center**, *Undergraduate Researcher*
Supervisor: Prof. Nuno Loureiro. Modeled particle transport in turbulent media using C.

2015 **MIT Department of Nuclear Science and Engineering**, *Undergraduate Researcher*
Supervisor: Prof. Emilio Baglietto. Modeled responses of nuclear waste storage canisters to fission waste using finite element software ADINA.

## Awards and Achievements

2020 – 2025 **JQI Graduate Fellowship**, Joint Quantum Institute, University of Maryland

2019, 2020 **NSF GRFP Honorable Mention**, National Science Foundation

## Press

2020 **PHYS.ORG**, *Popular science article*
New quantum computing algorithm skips past time limits imposed by decoherence, Online

## Service and Teaching

**Journal and Conference Reviews/Subreviews**
- Quantum Information Processing (QIP) 2023

2020 – 2022 **Volunteer Tutor**, *UMD Department of Physics*

2016 **Teaching assistant and grader**, *MIT Department of Physics*

## Publications

See also: Google Scholar, arXiv


[5] **J. T. Iosue**, K. Sharma, M. J. Gullans, and V. V. Albert, "Continuous-variable quantum state designs: theory and applications", 10.48550/arXiv.2211.05127 (2022).

[4] **J. T. Iosue**, A. Ehrenberg, D. Hangleiter, A. Deshpande, and A. V. Gorshkov, "Page curves and typical entanglement in linear optics", 10.48550/arXiv.2209.06838 (2022), *Submitted to PRX Quantum*.

[3] C. Cirstoiu, Z. Holmes, **J. T. Iosue**, L. Cincio, P. J. Coles, and A. Sornborger, "Variational Fast Forwarding for Quantum Simulation Beyond the Coherence Time", npj Quantum Information **6**, 82 (2020).

[2] P. Titum, **J. T. Iosue**, J. R. Garrison, A. V. Gorshkov, and Z.-X. Gong, "Probing ground-state phase transitions through quench dynamics", Physical Review Letters **123**, 115701 (2019).

[1] R. M. Parrish, **J. T. Iosue**, A. Ozaeta, and P. L. McMahon, "A Jacobi Diagonalization and Anderson Acceleration Algorithm For Variational Quantum Algorithm Parameter Optimization", 10.48550/arXiv.1904.03206 (2019).


## Talks

2022 **UC Boulder Journal Club**, *Invited*, Virtual
Continuous-variable quantum state designs: theory and applications (based on [5])

2022 **Quantum Seminar**, University of Maryland, College Park
Continuous-variable quantum state designs: theory and applications (based on [5])

2022 **APS March Meeting**, Virtual
Rigged continuous-variable $t$-designs (based on [5])

## Posters

2019 **TQC Conference**, College Park, Maryland
An initial condition robust outer-loop optimization strategy for a Quantum Approximate Optimization Algorithm

## Selected Projects

2019 – **qubovert**, *Python package for binary optimization*

- Created qubovert, which is particularly designed to aid in converting optimization problems to a form that can be solved with quantum annealers and quantum optimization algorithms.
- qubovert can be installed with `pip install qubovert`, the source code is hosted at `github.com/jtiosue/qubovert`, and the documentation is hosted at `qubovert.readthedocs.io`.
- qubovert currently has **over 191k downloads** from PyPI and 19 stars and 5 forks on GitHub.

2019 **Powell bounded multivariate optimization**, *SciPy contribution*

- Authored pull request number 10648 on Python's SciPy package. My contribution is included in the 1.5.0 release and later releases
- The pull request implements an additional feature for SciPy's minimization method. I devised a bounded version of the standard unbounded Powell minimization method and found it to often perform much better than the other gradient-free minimizers. I then implemented this variant in SciPy's software stack and created the pull request.

2018 **Quantum Computer Simulator**, *C++ project*

- Implemented a quantum computer simulator in C++.

## Skills

**Programming** Python, C, C++, Mathematica, Javascript, Julia, LaTeX, Git